Decision Trees

🎯 Classification Algorithm

📐 Entropy & Information Gain

🌳 ID3 Algorithm

What is a Decision Tree?

Root Node

The very first question asked. The single most informative feature in the dataset.

Internal Nodes

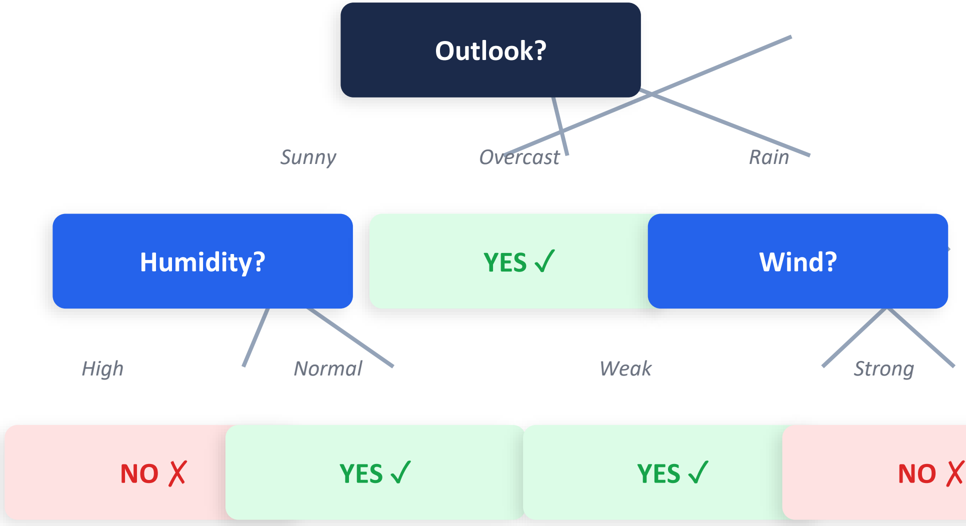
Intermediate decision points. Each tests one feature and splits the data based on its value.

Leaf Nodes

Terminal nodes at the bottom. Each holds a class label — the final prediction.

Branches

Edges connecting nodes. Each branch represents one possible outcome of its parent's test.



🌳 *A Decision Tree for Playing Tennis*

Entropy — Measuring Uncertainty

$$H(S) = -\Sigma\ p_i \cdot \log_2(p_i)$$

Pure Set → H = 0

All same class

{10 Yes, 0 No} → p = 1.0, 0.0

$H = -(1.0 \times \log_2 1.0) - 0 = 0$ bits

Zero uncertainty. No more questions needed!

Perfectly Mixed → H = 1.0

Equal classes — maximum uncertainty

{5 Yes, 5 No} → p = 0.5, 0.5

$H = -(0.5 \times -1) - (0.5 \times -1) = 1.0$ bit

Maximum uncertainty — coin flip!

Imbalanced → H = 0.88

More of one class

{7 Yes, 3 No} → p = 0.7, 0.3

$H = -(0.7 \times -0.515) - (0.3 \times -1.737) = 0.88$

Moderate uncertainty — some information.

💡 Key Insight: Entropy is 0 for pure sets and reaches maximum (1.0 bit) when classes are equally mixed.

Information Gain — Which Feature to Split On?

$$IG(S, A) = H(S) - \sum_{v \in Values(A)} (|S_v| / |S|) \cdot H(S_v)$$

Symbol Reference

$H(S)$	Entropy of the current set BEFORE the split
$Values(A)$	All possible values of attribute A
S_v	Subset of S where attribute A = value v
$\lvert S_v \rvert / \lvert S \rvert$	Fraction of examples with value v (the weight)
$H(S_v)$	Entropy of that subset AFTER the split

In Plain English

- IG = (Uncertainty BEFORE split) − (Weighted uncertainty AFTER split)
- HIGH IG → Split dramatically reduced uncertainty → GOOD split
- LOW IG → Split barely helped → BAD split
- ID3 always picks the attribute with the HIGHEST information gain
- The best question to ask is the one that tells you the MOST

💡 Algorithm Rule: At each node, compute IG for every remaining attribute. Choose the one with the highest IG as the split.

Tennis Dataset — Our Working Example

Day	Outlook	Temperature	Humidity	Wind	Play Tennis?
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
			Hig		

9 YES (Play)

5 NO (Skip)

14 Total Instances

4 Attributes

Step 1 — Compute Entropy of the Full Dataset H(S)

1

What is our starting uncertainty before any split?

Dataset: 14 examples

9 YES	5 NO
p = 9/14	p = 5/14

Formula	$H(S) = -(p\_yes \times \log_2 p\_yes) - (p\_no \times \log_2 p\_no)$
Substitute	$H(S) = -(9/14 \times \log_2 9/14) - (5/14 \times \log_2 5/14)$
Simplify	$H(S) = -(0.643 \times -0.637) - (0.357 \times -1.485)$
Calculate	$H(S) = 0.410 + 0.530$
Result	**$H(S) = 0.940$ bits**

📌 H(S) = 0.940 bits → We have significant uncertainty. The best possible split can reduce this toward 0. Our goal: find the attribute that reduces entropy the MOST.

Step 2 — Information Gain for OUTLOOK

SUNNY

Days: D1, D2, D8, D9, D11

2 YES | **3 NO**

$H = -(2/5 \log_2 2/5) - (3/5 \log_2 3/5)$

= 0.971 bits

Weight: 5/14

OVERCAST

Days: D3, D7, D12, D13

4 YES | **0 NO**

$H = -(4/4 \log_2 1) - 0$

= 0.000 bits ✨ PURE!

Weight: 4/14

RAIN

Days: D4, D5, D6, D10, D14

3 YES | **2 NO**

$H = -(3/5 \log_2 3/5) - (2/5 \log_2 2/5)$

= 0.971 bits

Weight: 5/14

$$IG(S, \text{Outlook}) = 0.940 - [\,(5/14 \times 0.971) + (4/14 \times 0.000) + (5/14 \times 0.971)\,]$$

$$= 0.940 - [\,0.347 + 0.000 + 0.347\,] = 0.940 - 0.693 = 0.246 \text{ bits}$$

✅ IG(S, Outlook) = 0.246 bits — Outlook will be compared with Humidity, Wind and Temperature next.

Step 3 — IG for Humidity, Wind & Temperature

HUMIDITY

High (D1,2,3,4,8,12,14)	7 ex	3Y / 4N	H = 0.985
Normal (D5,6,7,9,10,11,13)	7 ex	6Y / 1N	H = 0.592

`IG(S, Humidity) = 0.940 − 0.789 = 0.151`

WIND

Weak (D1,3,4,5,8,9,10,13)	8 ex	6Y / 2N	H = 0.811
Strong (D2,6,7,11,12,14)	6 ex	3Y / 3N	H = 1.000

`IG(S, Wind) = 0.940 − 0.892 = 0.048`

TEMPERATURE

Hot (D1,2,3,13) 4 examples, 2Y/2N, H=1.00	Mild (D4,8,10,11,12,14) 6 examples, 4Y/2N, H=0.918	Cool (D5,6,7,9) 4 examples, 3Y/1N, H=0.811

`IG(S, Temperature) = 0.940 − 0.911 = 0.029`

INFORMATION GAIN SUMMARY

Outlook **IG = 0.246 ← ROOT ✓**	Humidity IG = 0.151	Wind IG = 0.048	Temperature IG = 0.029

Step 4 — Split on OUTLOOK (Highest IG = 0.246)

OUTLOOK?

Sunny | *Overcast* | *Rain*

SUNNY Subset

Day	Temp	Hum	Wind	Play
D1	Hot	High	Weak	No
D2	Hot	High	Strong	No
D8	Mild	High	Weak	No
D9	Cool	Normal	Weak	Yes
D11	Mild	Normal	Strong	Yes

2 Yes / 3 No → H = 0.971

⚠ Not pure — need further split

OVERCAST Subset

Day	Temp	Hum	Wind	Play
D3	Hot	High	Weak	Yes
D7	Cool	Normal	Strong	Yes
D12	Mild	High	Strong	Yes
D13	Hot	Normal	Weak	Yes

4 Yes / 0 No → H = 0.000

✅ PURE LEAF! Always YES

RAIN Subset

Day	Temp	Hum	Wind	Play
D4	Mild	High	Weak	Yes
D5	Cool	Normal	Weak	Yes
D6	Cool	Normal	Strong	No
D10	Mild	Normal	Weak	Yes
D14	Mild	High	Strong	No

3 Yes / 2 No → H = 0.971

⚠ Not pure — need further split

Step 5 — Expand SUNNY Branch (Best Attribute: HUMIDITY)

Sunny subset: {D1, D2, D8, D9, D11} → 2 Yes, 3 No → H = 0.971 → Compute IG for remaining attributes

HUMIDITY ← WINNER ✓

High: {D1,D2,D8} → 0Y/3N H=0.000

Normal: {D9,D11} → 2Y/0N H=0.000

IG = 0.971 − 0 = 0.971

TEMP

Hot: {D1,D2} → 0Y/2N H=0.0

Mild: {D8,D11} → 1Y/1N H=1.0

Cool: {D9} → 1Y/0N H=0.0

IG = 0.971 − 0.4 = 0.571

WIND

Weak: {D1,D8,D9} → 1Y/2N H=0.918

Strong: {D2,D11} → 1Y/1N H=1.0

IG = 0.971 − 0.951 = 0.020

🏆 HUMIDITY wins with IG = 0.971 → Use HUMIDITY to split the Sunny branch

Humidity = HIGH → LEAF

D1, D2, D8 | 0 Yes / 3 No

Predict: NO ✗

Humidity = NORMAL → LEAF

D9, D11 | 2 Yes / 0 No

Predict: YES ✓

Step 6 — Expand RAIN Branch (Best Attribute: WIND)

Rain subset: {D4, D5, D6, D10, D14} → 3 Yes, 2 No → H = 0.971 → Compute IG for remaining attributes

HUMIDITY

High: {D4,D14}→1Y/1N H=1.0

Normal:{D5,D6,D10}→2Y/1N H=0.918

IG = 0.971 − 0.951 = 0.020

WIND ← WINNER ✓

Weak: {D4,D5,D10}→3Y/0N H=0.000

Strong: {D6,D14}→0Y/2N H=0.000

IG = 0.971 − 0 = 0.971

TEMP

Mild:{D4,D10,D14}→2Y/1N H=0.918

Cool:{D5,D6}→1Y/1N H=1.0

IG = 0.971 − 0.951 = 0.020

🏆 **WIND wins with IG = 0.971 → Use WIND to split the Rain branch**

Wind = WEAK → LEAF

D4, D5, D10 | 3 Yes / 0 No

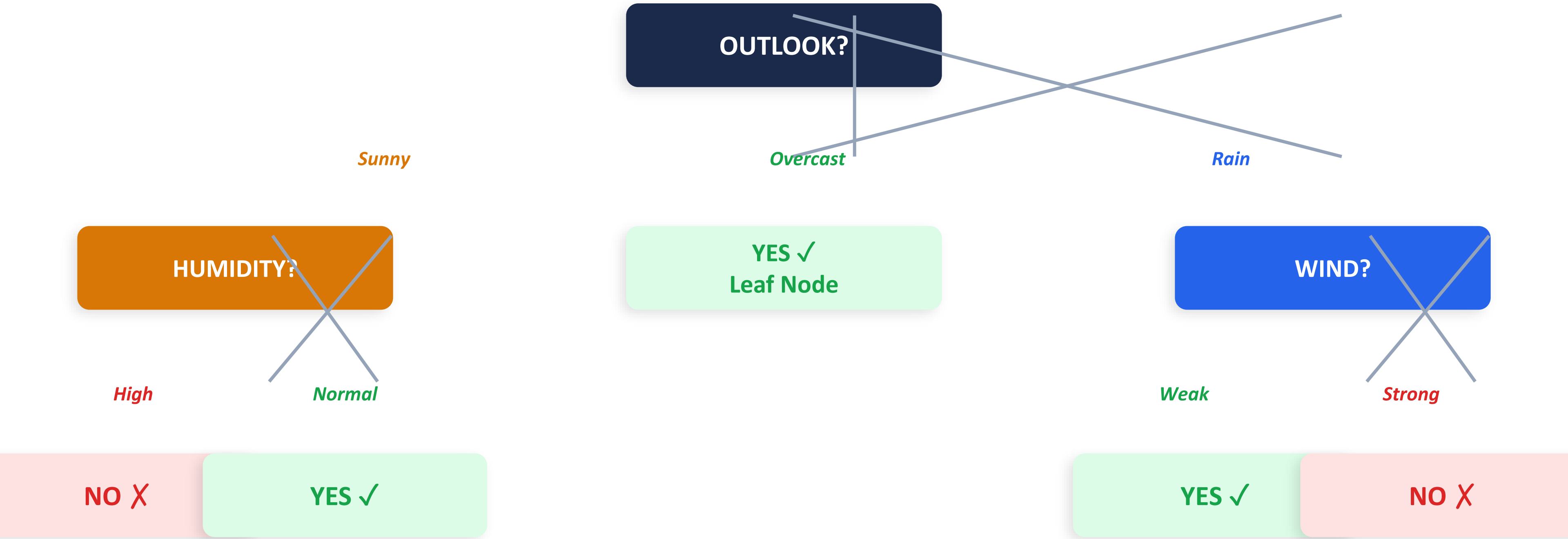
Predict: YES ✓

Wind = STRONG → LEAF

D6, D14 | 0 Yes / 2 No

Predict: NO ✗

🌳 The Complete Decision Tree — 100% Training Accuracy!



Extracted Classification Rules:

```
IF Outlook=Sunny AND Humidity=High      →  NO
IF Outlook=Sunny AND Humidity=Normal    →  YES
IF Outlook=Overcast                     →  YES
IF Outlook=Rain AND Wind=Weak           →  YES
IF Outlook=Rain AND Wind=Strong         →  NO
```

Advantages & Limitations of Decision Trees

ADVANTAGES

Interpretable
Rules can be read by doctors, lawyers, regulators. Each path is a human-readable rule.

No Scaling Needed
Unlike kNN, features don't need normalization. Trees use comparisons, not distances.

Mixed Data Types
Handles numeric, nominal, and ordinal features in the same model naturally.

Fast Prediction
O(depth) at test time — just follow the path. Microseconds for depth-10 tree.

Implicit Feature Selection
Features never appearing in the tree are irrelevant — built-in importance measure.

LIMITATIONS

High Variance
Small data changes can produce completely different trees — unstable.

Axis-Aligned Only
Cannot directly model diagonal boundaries. Needs many splits to approximate.

Greedy — Not Optimal
ID3 picks locally best attribute. Finding globally optimal tree is NP-hard.

Overfits Easily
Without pruning: 100% training, poor generalization. Pruning is mandatory.

No Extrapolation
For regression: can only predict values seen in training. Cannot go beyond range.

Key Takeaways

1 **Decision Trees ask sequential questions**

Root → Internal Nodes → Leaves. Each path from root to leaf is one classification rule.

2 **Entropy measures uncertainty**

$H(S) = -\Sigma\ p_i \log_2(p_i)$. Zero for pure sets, maximum (1.0 bit) for equal class mix.

3 **Information Gain selects the best split**

IG = H(S) − weighted average H of subsets. ID3 always picks the highest IG attribute.

4 **Overfitting is the greatest danger**

Full trees memorize noise. Pre-pruning or post-pruning is mandatory for good generalization.

5 **Trees are the foundation of modern ML**

Random Forests and Gradient Boosting (XGBoost) are ensembles of Decision Trees — dominant on tabular data.